



ECDS



DATAQUEST 3.0

DQPS

Title

Next-Generation Autonomous D2C Advertising Intelligence & Decision Engine

Background

Design and build a 0-to-1 AI-native advertising intelligence and autonomous decision-making system for Direct-to-Consumer (D2C) brands, ensuring that it satisfies both:

1. Unified Cross-Platform Intelligence — Ingesting, reconciling, and reasoning across disparate ad platforms, sales channels, pricing, margins, inventory, and creative assets.
2. Autonomous Decisioning & Closed-Loop Learning — Moving beyond passive analytics dashboards to proactive diagnosis, predictive opportunity scoring, and automated execution feedback loops.

D2C advertising is becoming exponentially complex. Brands operate across multiple fragmented advertising networks (Meta, Google, Amazon, TikTok, programmatic), product catalogs, target audiences, creative variations, dynamic pricing models, and fluctuating inventory levels. Existing tooling merely collects or visualizes historical numbers in passive dashboards. Brand managers are overwhelmed by data silos and struggle to determine which campaign to adjust, why performance suddenly spiked or dropped, and where the next dollar will yield maximum profitable growth. The goal is to create an AI-native system that understands what is happening, diagnoses root causes, predicts winning opportunities before spend occurs, and autonomously drives profitable growth.

Problem

D2C brands managing multi-channel advertising operations often suffer from:

- Siloed performance data: Advertising spend, sales revenue, product margins, inventory levels, and creative metrics reside in disconnected systems.

- Manual and delayed diagnosis: Identifying the root cause of performance fluctuations requires tedious cross-channel manual correlation.
- Reactive and intuition-based spend: Budget allocation across products, audiences, and ad creatives relies on guesswork rather than predictive intelligence.
- Suboptimal marginal return: Inability to dynamically reallocate budgets towards high-margin, high-inventory SKUs with highest conversion probability.
- Lack of decision automation: Analytics tools present descriptive charts but fail to generate executable decisions or measure the impact of actions taken.

These bottlenecks lead to wasted ad spend, stockouts on promoted items, compressed profit margins, and missed revenue opportunities. We require an AI-native decision engine that reconciles multi-source data, reasons over business constraints, predicts optimal actions, and continuously learns from outcome feedback.

Challenge

Design an intelligent advertising decision system that analyzes multi-channel marketing signals to guide profitable growth, ensuring that it satisfies both:

1. Cross-channel data synthesis and diagnostic reasoning
2. Actionable decision generation and outcome learning

1. System Scope:

- Ingestion and correlation of advertising spend, sales, inventory, and margin metrics.
- Identification of performance anomalies and primary contributing factors.
- Formulation of prioritized budget and campaign adjustment recommendations.

2. Decision Scenario:

Scenario:

Evaluating performance fluctuations across active campaigns. The system should:

- Detect significant shifts in advertising efficiency or conversion patterns.
- Surface probable drivers behind observed changes across product or campaign lines.
- Recommend high-impact budget allocations aligned with business objectives.

Core Concepts

The core system orchestrates an autonomous closed-loop intelligence lifecycle:

- Ingest and Reconcile: Continuously aggregate multi-platform ad spend, eCommerce sales, Google Analytics events, ERP inventory, and SKU margin data into unified schema vectors.
- Diagnose and Reason: AI agents and causal inference models detect anomalies, isolate root causes, and reason across profitability constraints.

- Decide, Execute, and Learn: Generate high-confidence actionable directives, execute approved changes via ad APIs, and continuously reinforce models via outcome feedback.

Wishing all participating teams the very best of innovation and success at DataQuest 3.0!